Rohan Malhotra

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EXPERIENCE

IBM Dallas, TX

Robotics Engineer Intern *Jun 2026 – Aug 2026*

- Built a real-time pipeline connecting YOLO11 inference to the Boston Dynamics Spot SDK via OpenCV and gRPC.
- Designed a multithreaded service with lock-free inference queues; demoed live retrieval at IBM DevCon.
- Open-sourced weights, training pipelines, and docs; IBM sales and client-success teams adopted the demo.

Applications Developer Intern, IBM Consulting *May 2026 – Aug 2026*

- Built an on-prem EPM assistant that turns plain-language requests into reviewable JSON using RAG and MCP tools.
- Connected banking APIs to ERP and EPM via Oracle Integration Cloud and migrated Excel workflows to Oracle Cloud.
- Fine-tuned Qwen-Coder-32B on Oracle EPM assets, raising EPM build-generation accuracy from **36.7% to 95.0%**.

Kalshi New York, NY

Quantitative Developer *Jan 2026 – May 2026*

- Built a C++ hedging engine that converts contract prices (cents, probability, bps) into real-time exposure.
- Architected a WebSocket market-data pipeline ingesting quotes and trades to drive live pricing and hedge signals.
- Built a Next.js/Python engine that recommends hedge size and compares cost vs. tail risk using FRED and BLS data.
- Wrote C++ risk tools, containerized the runtime with Docker, and deployed it to Render.

Hume Center for National Security and Technology, Virginia Tech Blacksburg, VA

Aerospace Researcher *Nov 2024 – May 2025*

- Programmed imaging and signal-processing tests in C for ContentCube, a CubeSat deployed from the ISS in 2025.
- Engineered a test environment simulating concurrent imaging workloads to resolve race conditions and memory bugs.

Aress Software New York, NY

Machine Learning Intern *Jun 2025 – Aug 2025*

- Trained and deployed PyTorch models on **500K+ multimodal records** across Azure and AWS.
- Designed NoSQL pipelines unifying 10+ data sources into one analytics layer; optimized SQL queries for speed.

PROJECTS

NightShift | *FastAPI, Playwright, Next.js, Stripe* 2026

- Built a platform that finds jobs, fills applications via browser automation, and verifies confirmation emails.
- Added scheduled runs, Stripe subscriptions, and a live dashboard with a full application log.

Refrax | refrax.io | *C++, Next.js, TypeScript, PostgreSQL* Jun 2025 – Feb 2026

- Designed low-latency REST/WebSocket pipelines streaming market data into a PostgreSQL research environment.
- Implemented C++ risk engines (Greeks, P&L simulation, portfolio exposure) optimized for parallel execution.

EPM Wizard | *FastAPI, React, Groovy, Playwright* 2026

- Built a tool that turns plain-English requests into Oracle EPM forms and rules, with preview and approval steps.

IndyTrack API | *Flask, SQLAlchemy, PostgreSQL* 2025

- Built a task-tracker backend with user accounts, due dates, alerts, notifications, and SQL persistence.

EDUCATION

New York University, Courant Institute of Mathematical Sciences New York, NY

B.A. in Computer Science, Minor in Mathematics *Expected May 2027*

- **GPA:** 3.7/4.0 | Accelerated three-year degree | **Graduate course:** Mathematical Techniques for CS Applications
- **Coursework:** Data Structures, Algorithms, Operating Systems, Computer Organization, Software Engineering

SKILLS

Languages: Python, C++, C, Java, TypeScript, JavaScript, SQL, Rust, Groovy
Frameworks: React, Next.js, FastAPI, Flask, Node.js, Playwright, PyTorch, OpenCV
Infrastructure: Linux, Git, Docker, Kubernetes, AWS, Azure, PostgreSQL, gRPC, WebSockets, REST APIs